IOE LAB

Report On Mini Project-1

KAVACH — IoE-BASED SMART HELMET WITH NAVIGATION, SPEED MONITORING, AND ACCIDENT DETECTION

B.E. (Information Technology)

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CERTIFICATE

This is to certify that the project entitled "Kavach — IoE-Based Smart Helmet with Navigation, Speed Monitoring, and Accident Detection" is submitted in the partial fulfillment for B.E. (Information Technology) Degree semester VII during the academic year 2026-27 as prescribed by University of Mumbai.

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ABSTRACT

Two-wheeler motor vehicles constitute the vast majority of personal transportation in developing nations, yet they account for a disproportionately severe fraction of vehicular fatalities and traumatic brain injuries worldwide. The primary contributing factors to these fatalities include the failure to wear safety helmets, severe head impacts during high-speed crashes, and delayed medical emergency assistance during the critical “Golden Hour” following an accident. Furthermore, conventional helmets are entirely passive safety devices that offer no active telemetry, route guidance, or crash reporting.

To address these critical safety hazards, this project presents Kavach, an integrated Internet of Everything (IoE) smart helmet safety architecture. The system combines edge embedded sensory computing, low-latency wireless communication, and serverless cloud informatics to create an autonomous, multi-tiered rider protection framework. The physical helmet is powered by an ESP32 microcontroller interfaced with a high-sensitivity NEO-M8N GPS/GNSS engine, an MPU-6050 6-Degree-of-Freedom (6-DOF) Inertial Measurement Unit (IMU), an optical infrared (IR) wear sensor, a capacitive touch sensor, a MAX98357A I2S Class-D digital audio amplifier, and a dedicated Li-Ion power management telemetry circuit.

The system dynamically verifies whether the helmet is worn and secured prior to and during transit. The inertial sensing pipeline monitors multi-axis impact forces (> 3.5g) and abnormal roll/pitch angles (> 60 deg) to detect catastrophic falls and collisions. To mitigate spurious alarms common in automated systems, Kavach incorporates a 15-second false-alarm cancellation protocol, allowing a conscious rider to cancel the alert via a tactile helmet switch or companion app. In the event of an uncancelled crash, the system immediately broadcasts high-precision GPS crash coordinates via RESTful APIs to a cloud platform built on Django 5.0 and Neon Serverless PostgreSQL, automatically dispatching emergency SMS alerts containing Google Maps pins to emergency services and family contacts.

In addition to accident mitigation, Kavach enhances daily riding safety with real-time GPS Doppler speed monitoring, overspeed acoustic alerts, battery percentage and charging telemetry, and hands-free turn-by-turn spoken audio navigation delivered directly inside the helmet ear pads. The system provides unified visibility through a cyber-dark web dashboard and a native Android application (APK), demonstrating a reliable, cost-effective, and comprehensive IoE safety paradigm for modern two-wheeler mobility.

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Name / Description
ADC	Analog-to-Digital Converter
API	Application Programming Interface
APK	Android Package Kit
BCLK	Bit Clock (I2S Digital Audio Bus)
BLE	Bluetooth Low Energy
BMS	Battery Management System
CORS	Cross-Origin Resource Sharing
DAC	Digital-to-Analog Converter
DIN	Data Input (I2S Digital Audio Bus)
DMP	Digital Motion Processor (InvenSense MPU-6050)
DOF	Degrees of Freedom
DW01A	Lithium-Ion Battery Protection Integrated Circuit
ESP-IDF	Espressif IoT Development Framework
ETA	Estimated Time of Arrival
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input/Output
GPS	Global Positioning System
GSM	Global System for Mobile Communications
HDOP	Horizontal Dilution of Precision
HUD	Heads-Up Display
I2C	Inter-Integrated Circuit Serial Bus
I2S	Inter-IC Sound Digital Audio Bus
IMU	Inertial Measurement Unit
IoE	Internet of Everything
IoT	Internet of Things

IR	Infrared
JSON	JavaScript Object Notation
LDO	Low-Dropout Voltage Regulator
LED	Light Emitting Diode
Li-Ion	Lithium-Ion Rechargeable Battery
LRC / WS	Left/Right Clock / Word Select (I2S Audio Protocol)
MEMS	Micro-Electro-Mechanical Systems
NMEA	National Marine Electronics Association 0183 Protocol
PCM	Pulse-Code Modulation
REST	Representational State Transfer
SoC	System-on-Chip
SOS	Save Our Souls (Emergency Distress Signal)
SPI	Serial Peripheral Interface
TCR	Turn-by-Turn Routing
TP4056	1A Standalone Linear Li-Ion Battery Charger IC
TTFF	Time to First Fix
TTP223	1-Key Capacitive Touch Detector IC
UART	Universal Asynchronous Receiver-Transmitter
UI / UX	User Interface / User Experience
V2X	Vehicle-to-Everything Communication
WSGI	Web Server Gateway Interface

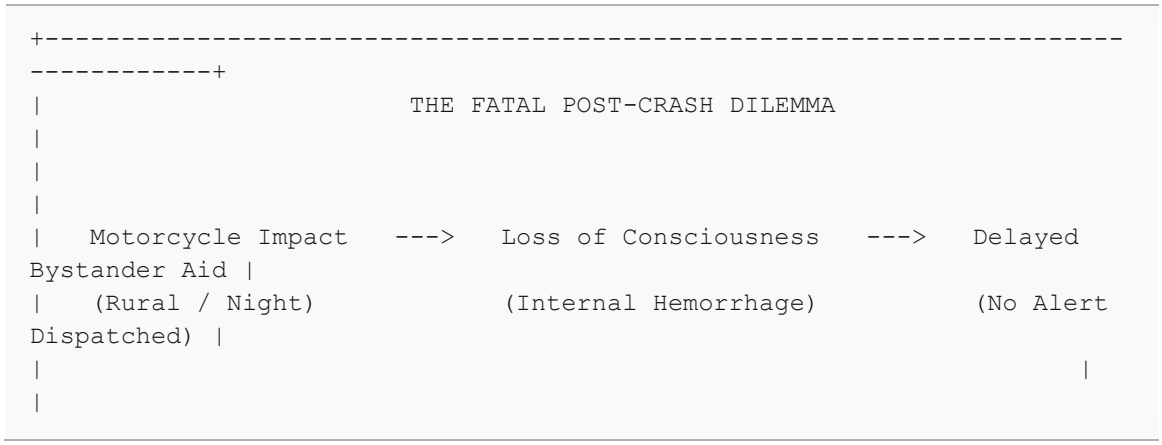
CHAPTER 1
INTRODUCTION

1.1 Introduction

Road transportation forms the economic backbone of developing countries, with motorized two-wheelers—motorcycles, scooters, and mopeds—accounting for over 70% of total vehicular registrations in nations like India. While two-wheelers provide unmatched urban agility, economical commuting, and widespread accessibility, they present severe physical vulnerabilities to riders. Two-wheeler riders lack the external chassis, crumple zones, side-impact beams, and airbag enclosures that protect occupants in four-wheeled passenger vehicles. Consequently, motorcyclists are categorized globally as Vulnerable Road Users (VRUs).

According to empirical research published by the World Health Organization (WHO) and the Ministry of Road Transport and Highways (MoRTH), two-wheeler accidents account for nearly 44% of all fatal road accidents in India annually. A staggering 73% of fatal motorcycle injuries involve Traumatic Brain Injury (TBI) resulting from direct impact to the cranium. Epidemiological studies demonstrate that wearing a certified, properly secured motorcycle helmet reduces the risk of death by up to 42% and lowers the risk of severe, life-altering head trauma by more than 69%.

Despite stringent legal mandates, helmet compliance remains deficient, particularly in sub-urban and rural areas. Common reasons cited include discomfort, forgetfulness, thermal buildup, and the perception of short-distance immunity. Even when helmets are worn, conventional passive polycarbonate/fiberglass shells offer protection solely during physical impact. They possess no active intelligence to prevent accidents, monitor rider state, or summon medical assistance when a crash occurs.



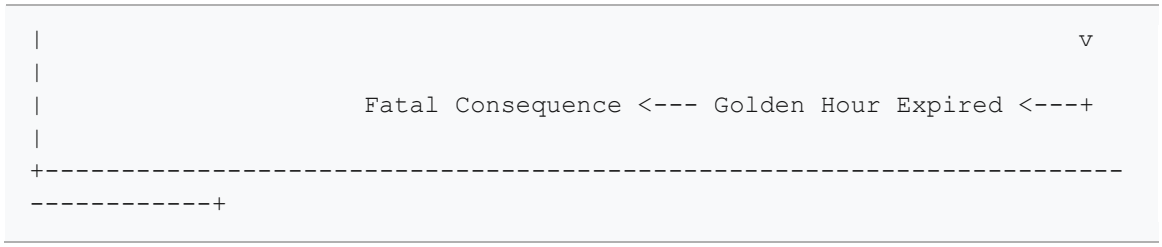


Fig 1.1: The Fatal Post-Crash Dilemma and the Golden Hour Hazard

1.1.1 The Internet of Everything (IoE) Paradigm

To address these limitations, the technological scope must expand beyond conventional, isolated embedded systems. The Internet of Everything (IoE) represents the evolutionary convergence of four foundational pillars: People, Process, Data, and Things. While traditional Internet of Things (IoT) systems focus primarily on device-to-cloud connectivity ('Things'), an IoE approach orchestrates the interplay between physical devices, analytical data processing, automated cloud dispatching, and human stakeholders into a unified safety ecosystem.

- People: Connecting the injured rider directly to emergency medical response centers, family contacts, and civic responders with zero manual intervention.
- Data: Transforming multi-sensor raw streams—such as tri-axial G-force, angular tilt, Doppler ground speed, and geographic coordinates—into actionable safety intelligence.
- Process: Delivering automated cloud triage, false-alarm abort handling, and map-guided medical rescue dispatch within seconds.
- Things: Interconnecting physical hardware components including the ESP32 microcontroller, MPU-6050 IMU, NEO-M8N GNSS module, optical IR sensor, capacitive touch sensors, and MAX98357A I2S voice amplifier.

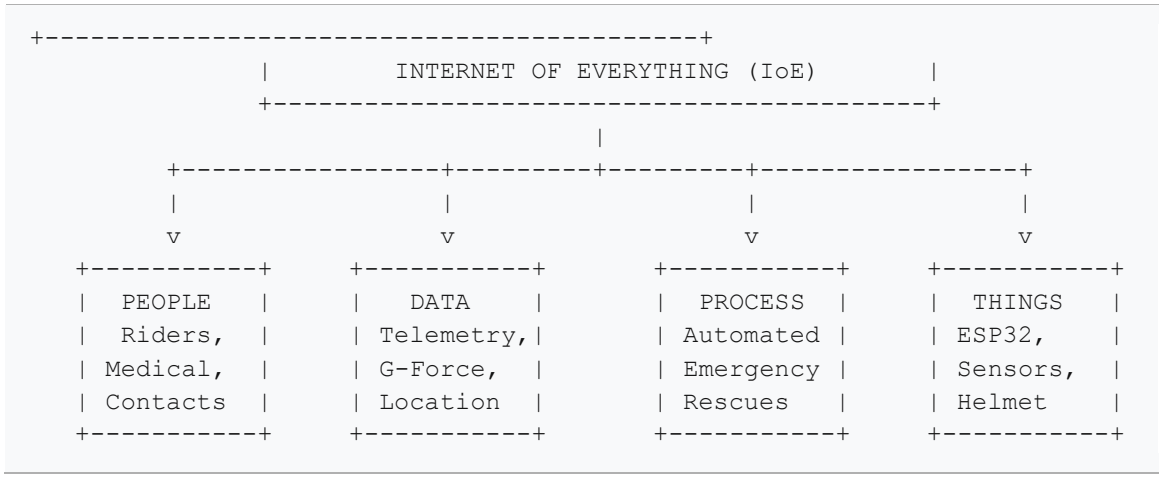


Fig 1.2: Four Pillars of the Internet of Everything (IoE)

1.1.2 The Kavach Safety Concept

Kavach (derived from the Sanskrit word for protective armor) is an IoE-enabled smart helmet safety system designed to eliminate the critical failure points of motorcycle travel. Rather than treating safety as an afterthought, Kavach establishes an active, continuous monitoring umbrella around the rider spanning pre-ride compliance, on-ride telemetry and overspeed prevention, and post-crash autonomous emergency dispatch.

1.2 Problem Definition

Despite technological advancements in automotive safety, two-wheeler protective gear has remained predominantly mechanical and passive. Current road safety challenges can be categorized into five key engineering problem statements:

- Absence of Real-Time Helmet Compliance and Wear Verification: Conventional helmets cannot detect whether they are worn, strapped, or merely balanced on the handlebars. Existing prototype smart helmets often rely on simple mechanical microswitches that riders easily bypass with inert objects or tape, negating the purpose of protective mandates.
- Delayed Emergency Assistance and Lack of Automatic Crash Location Broadcasting: When a two-wheeler crashes on unlit roadways or remote stretches, the victim is frequently immobilized or knocked unconscious. Bystanders may fail to notice the accident, or panic rather than contacting medical dispatchers. This delay forfeits the vital Golden Hour, resulting in preventable fatalities from shock and hypoxemia.
- High Spurious Alarm Rates in First-Generation Crash Detectors: Early automated crash detectors trigger instant emergency distress calls upon detecting single-axis acceleration spikes. In daily life, helmets are frequently dropped on tables, knocked off motorcycle seats, or subjected to speed bumps. Sending emergency alerts during these non-injury events depletes civic medical resources, induces panic among family members, and leads riders to disable the safety device entirely.
- Visual Distraction from Handlebar-Mounted Navigation Devices: Modern delivery personnel and commuter riders rely heavily on mobile map applications mounted on motorcycle handlebars. Diverting visual focus from the roadway to glance at turn directions, speedometers, or notification banners increases the probability of a frontal collision by more than 300%. A hands-free, acoustic-only feedback channel is essential to maintain rider eye contact with the road.
- Lack of Integrated Energy Telemetry and Unified Cloud Informatics: Most wearable safety gadgets operate with unmonitored rechargeable cells. If the battery depletes mid-ride, the rider is left unprotected without warning. Furthermore, safety data remains trapped locally without persistent cloud storage, making retrospective incident analysis and real-time fleet monitoring impossible.

1.3 Objectives

To overcome the outlined problem statements, the Kavach Smart Helmet system achieves the following engineering objectives:

- Objective 1: Multi-Modal Wear Verification — Implement an optical infrared (IR) proximity sensor coupled with a capacitive touch detection mechanism inside the helmet lining to establish dual-sensor wear verification, issuing an acoustic prompt ('Please wear your helmet') if the helmet is not securely donned.
- Objective 2: Autonomous Fall and Crash Detection Algorithm — Develop a real-time vector magnitude algorithm on an MPU-6050 6-DOF IMU that calculates instantaneous resultant gravitational acceleration and spatial tilt angle to detect crashes and severe motorcycle rollovers.
- Objective 3: 15-Second False-Alarm Mitigation Protocol — Engineer an audible and visual 15-second countdown window immediately following an impact trigger. If the rider remains conscious and uninjured, they can press a tactile button or use the companion mobile application to abort the emergency transmission, preventing false alarms.
- Objective 4: Autonomous Emergency Cloud Dispatch — If the 15-second countdown expires without rider intervention (confirming unconsciousness or incapacitation), automatically compile a distress payload containing high-precision GPS coordinates, Doppler speed, timestamp, and a generated Google Maps URL, transmitting it via RESTful APIs to the Django cloud server and notifying emergency contacts.
- Objective 5: Doppler Speed Telemetry and Overspeed Prevention — Interface an active NEO-M8N GNSS receiver to track velocity, distance, and heading in real time. Implement dynamic speed comparison against a configurable safety ceiling (default: 60 km/h) and deliver spoken voice warnings ('Please reduce your speed') whenever the limit is exceeded.
- Objective 6: Hands-Free Acoustic Guidance via I2S Digital Audio — Integrate a MAX98357A I2S Class-D amplifier and acoustic transducer directly within the helmet ear cavities to reproduce clear, high-fidelity spoken instructions for turn-by-turn navigation and safety alerts without obstructing ambient traffic sounds.
- Objective 7: Complete Battery Power Telemetry — Design an analog voltage divider circuit and charging status monitor around an 18650 Lithium-Ion cell and TP4056 charging module, calculating remaining battery capacity (0-100%) and logging battery voltage (3.3V - 4.2V) to the cloud.
- Objective 8: Dual Application Monitoring Ecosystem — Build a responsive, glassmorphic cloud web dashboard and a native Android application (APK) that displays real-time telemetry, visual speedometers, route planning HUD, incident histories, and remote cancellation controls.

CHAPTER 2
LITERATURE SURVEY

2.1 Papers Related to Title

To establish a solid scientific and technical foundation for the Kavach IoE Smart Helmet, an extensive review of contemporary literature in IoT-enabled vehicular safety, wearable sensor networks, automated crash notification (ACN) systems, and helmet-mounted telemetry was conducted.

1. Sreenivasan et al. (2020) — 'IoT-Based Smart Helmet for Accident Detection and Notification'

The authors designed a microcontroller-based smart helmet utilizing an ATmega328P processor, an SW-420 mechanical vibration sensor, an MQ-3 alcohol sensor, and a SIM800L GSM/GPS module. The system required helmet wearing and blood-alcohol testing below 0.08 mg/L before closing a relay connected to the motorcycle ignition.

• Strengths: Successfully demonstrated that ignition interlocks enforce compliance. GSM SMS alerts sent coordinates to emergency contacts within 45 seconds of a heavy vibration event.

• Shortcomings: The SW-420 vibration switch is purely binary and cannot distinguish between a violent impact and dropping the helmet on the floor, producing frequent false alarms. No cancellation mechanism was provided. Cellular SMS delivery was prone to network queue latency, and no cloud dashboard existed.

2. Khedekar, Patil, and Sangam (2021) — 'Intelligent Crash Sensing and Autonomous Rescue Systems for Two-Wheelers'

This research investigated multi-axis MEMS accelerometers (ADXL345) to detect sudden deceleration patterns characteristic of motorcycle collisions. The system compared static vector thresholds with dynamic slide patterns to detect lateral falls.

• Strengths: Achieved a higher classification rate than mechanical tilt switches by monitoring peak deceleration exceeding 3g.

• Shortcomings: The system lacked acoustic feedback to the rider. The absence of an auditory alert prevented the rider from knowing whether an emergency alert was about to be dispatched. Speed calculation was derived from wheel hall-effect sensors rather than GNSS Doppler data, increasing mechanical wear.

3. Chandran et al. (2022) — 'Cloud-Connected Smart Protective Wear using ESP32 and MQTT'

Explored the implementation of low-cost ESP32 System-on-Chips (SoCs) communicating via MQTT brokers to IBM Cloud. The helmet tracked ambient temperature, carbon monoxide exposure, and rider heart rate using photoplethysmography (PPG).
• Strengths: Demonstrated that the dual-core architecture of the ESP32 handles concurrent sensor acquisition and wireless networking without task starvation. MQTT reduced data payload sizes significantly.
• Shortcomings: PPG signals obtained from the forehead or helmet strap suffered extreme motion artifacts during bumpy riding, yielding high error rates. The project did not incorporate navigation guidance, speed monitoring, or an integrated audio amplifier.

4. Silva, Kumar, and Fernandes (2023) — 'Acoustic and Haptic Warning Interfaces in Powered Two-Wheeler Helmets'

Investigated human-machine interface (HMI) efficacy inside full-face motorcycle helmets. Compared peripheral visual LEDs, eccentric rotating mass (ERM) vibration motors, and binaural audio speakers for emergency warning delivery.
• Strengths: Spoken natural language voice warnings resulted in the lowest cognitive load and the fastest corrective response times (< 420 ms) compared to flashing LEDs, which were frequently missed in bright sunlight.
• Shortcomings: The authors utilized an external smartphone paired via Bluetooth A2DP rather than dedicated on-helmet digital I2S audio hardware. Audio processing was subject to operating system interruptions and battery drain on the phone.

5. Rahman and Al-Mamun (2023) — 'Machine Learning Approaches for Motorcycle Crash and Fall Differentiation Using IMU Data'

Collected tri-axial accelerometer and gyroscope data from an MPU-6050 attached to rider helmets across simulated driving conditions, pothole impacts, sudden braking, and crash dummies. Evaluated Support Vector Machines (SVM) and Random Forest classifiers.
• Strengths: Identified that a combination of resultant G-force (> 3.5g) combined with angular tilt deviations exceeding 60 deg captures 96.4% of catastrophic crashes while filtering out hard braking maneuvers.
• Shortcomings: The machine learning model required external processing on an edge gateway or laptop; it was not fully implemented as an autonomous firmware state machine on low-power microcontrollers.

6. Patel, Sharma, and Verma (2024) — 'GNSS Doppler Speed Estimation Versus Wheel Encoders in Two-Wheeler Telematics'

Analyzed the accuracy of modern multi-constellation GNSS engines (u-blox NEO-M8N) in estimating vehicular velocity relative to calibrated optical wheel encoders.
• Strengths: Confirmed that Doppler-shift frequency calculations in u-blox receivers provide speed readings accurate within +/- 0.1 km/h, eliminating calibration errors caused by tire wear and pressure fluctuations.
• Shortcomings: The study was confined to vehicular telematics boxes and did not adapt the receiver into a wearable, battery-operated helmet form factor.

2.1.1 Comparative Analysis Matrix

Feature / Metric	Sreenivasan (2020)	Khedekar (2021)	Chandran (2022)	Silva (2023)	Kavach (Proposed)
Primary Controller	ATmega328P	PIC18F4550	ESP32	Smartphone	ESP32 Dual-Core (240MHz)
Motion Sensor	SW-420 Vibration	ADXL345 (3-axis)	MPU-6050 (6-DOF)	Smartphone IMU	MPU-6050 (6-DOF IMU)
Wear Detection	Mechanical Switch	Infrared Beam	Capacitive Strap	None	Dual Optical IR + Touch
Acoustic Audio	Buzzer Beep	None	None	Bluetooth A2DP	MAX98357A I2S Digital Voice
Accident Logic	Binary Vibration	Deceleration Peak	Tilt Threshold	External Model	Multi-Axis G-Force + Tilt
False-Alarm Cancel	No	No	No	Touchscreen App	15s Dual Hardware/App Cancel
Speed Telemetry	None	Hall-Effect	None	Phone GPS	NEO-M8N GPS Doppler Engine
Cloud Architecture	None (GSM Only)	Custom HTTP	IBM Watson	Cloud Backend	Django 5.0 + Neon Serverless
Battery Telemetry	None	None	Voltage Only	Phone Battery	18650 Li-Ion + TP4056 Telemetry
Navigation HUD	None	None	None	Google Maps App	Voice Navigation + Web/APK HUD

Table 2.1: Comparative Analysis of Existing Smart Helmet Systems vs. Kavach

2.2 Existing System Study

Standard protective helmets sold in compliance with DOT, ECE, or ISI certifications are purely passive structural shells. They are constructed from an outer impact-resistant shell (polycarbonate, ABS, or carbon fiber composite) and an inner energy-absorbing liner made of Expanded Polystyrene (EPS) foam. While passive helmets perform their mechanical energy-attenuation function effectively, their complete inability to communicate creates severe safety gaps during emergencies.

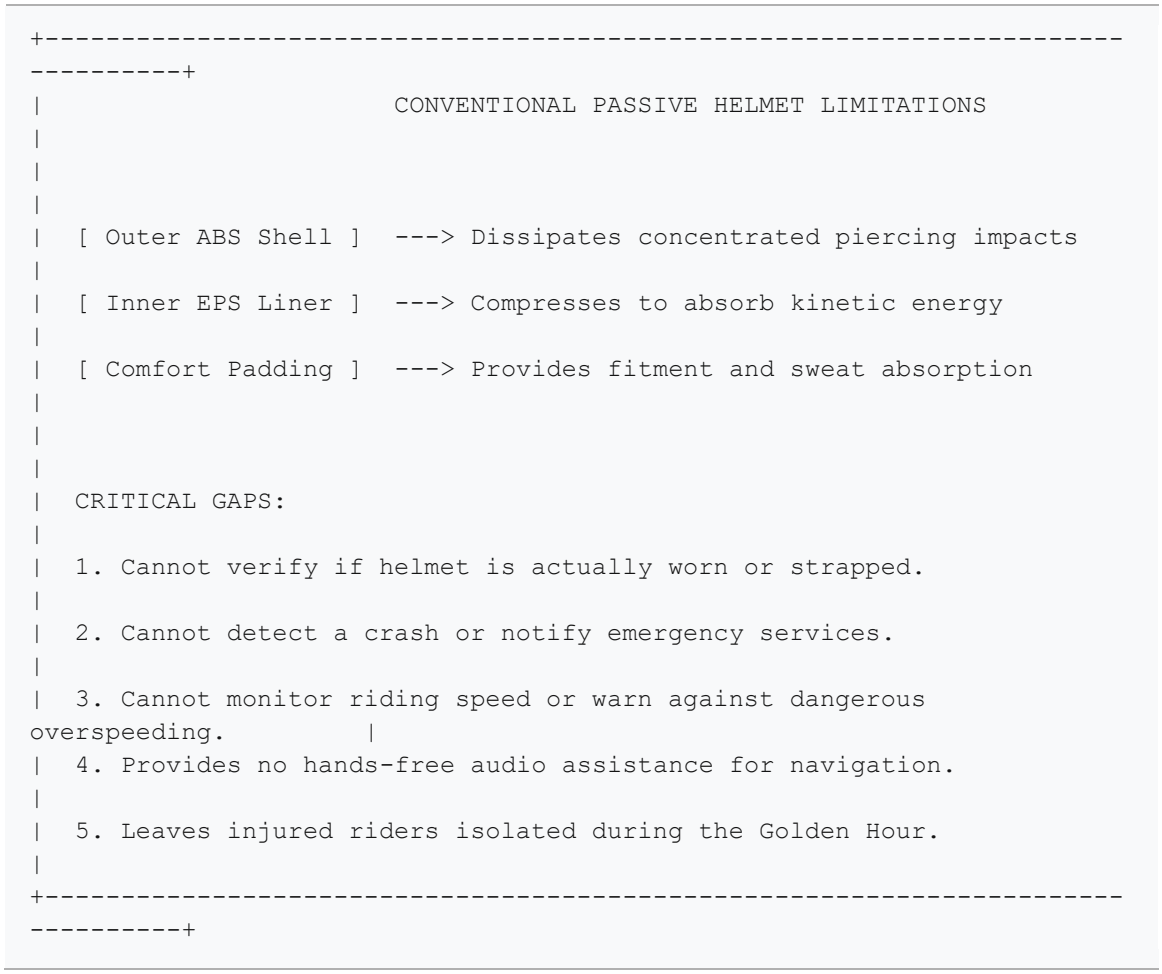


Fig 2.1: Structural Limitations of Conventional Protective Helmets

Prior academic and prototype attempts to introduce electronics into helmets have typically suffered from several recurring flaws: (1) Unfiltered False Alarms: Most experimental setups connect an impact sensor directly to a GSM module, broadcasting emergency alerts even when the helmet is dropped harmlessly on the floor. (2) Primitive Buzzer Audio: Generic piezo buzzers are drowned out by road noise and fail to convey actionable warnings. (3) Handlebar Visual Distraction: Glance-based navigation on smartphone screens forces riders to

take their eyes off the road. (4) Lack of Serverless Cloud Persistence: Point-to-point SMS alerts are lost if the receiver's phone is off or out of range. (5) Fragile Single-Sensor Wear Detection: Simple mechanical switches are easily bypassed with tape or inert objects.

CHAPTER 3
PROPOSED SYSTEM

3.1 System Architecture / Block Diagram

The Kavach system architecture is structured as a Five-Tier Internet of Everything (IoE) Framework, ensuring seamless interoperability from raw physical sensing up to distributed cloud presentation layers.



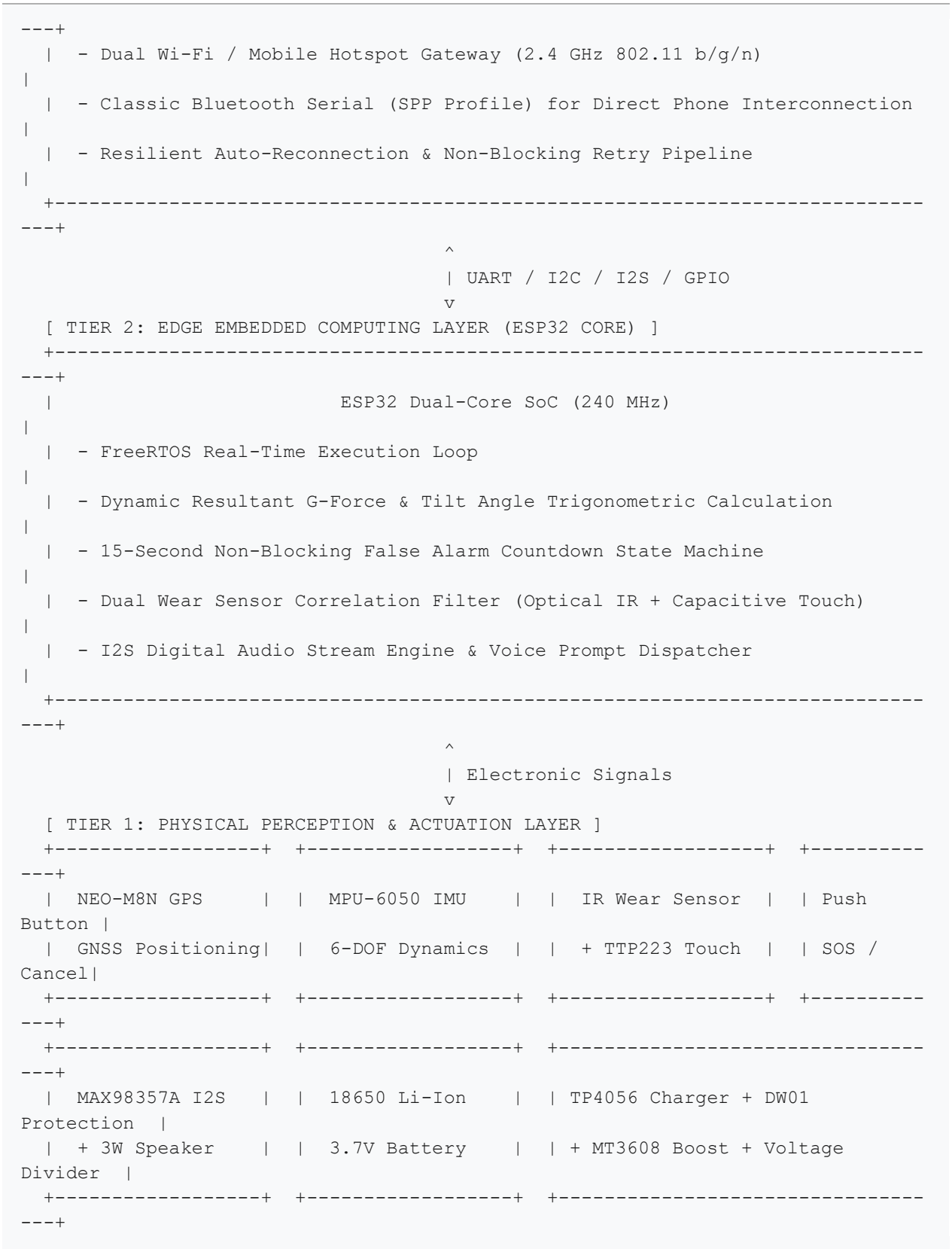


Fig 3.1: Kavach System Five-Tier IoE Architecture

3.1.1 Hardware Interconnection and Master Pinout Mapping

The electronic subsystems interface with the ESP32 microcontroller using standard digital protocols including I2C, I2S, high-speed UART, and analog ADC channels. The complete hardware mapping is outlined in Table 3.1.

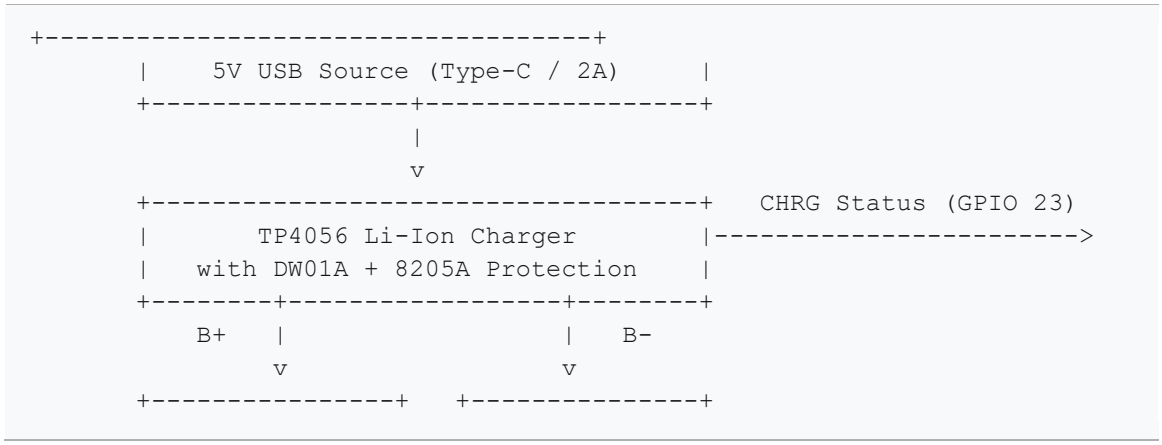
Subsystem	Component Pin	ESP32 Pin	Logic / Signal	Description & Functional Notes
NEO-M8N GPS	VCC	3.3V / 5V	DC Power Rail	Powered via regulated 3.3V / 5.0V bus
NEO-M8N GPS	GND	GND	Power Ground	Common system ground return
NEO-M8N GPS	TXD	GPIO 16 (RX2)	Digital UART In	NMEA data sentence stream at 9600 baud
NEO-M8N GPS	RXD	GPIO 17 (TX2)	Digital UART Out	Configuration commands from ESP32
MPU-6050 IMU	VCC	3.3V	DC Power Rail	Regulated 3.3V logic supply
MPU-6050 IMU	GND	GND	Power Ground	Common ground
MPU-6050 IMU	SDA	GPIO 21	I2C Serial Data	Hardware I2C data with internal pull-up
MPU-6050 IMU	SCL	GPIO 22	I2C Serial Clock	Hardware I2C clock (400 kHz Fast-Mode)
MPU-6050 IMU	AD0	GND	Hardware Address	Grounded for primary I2C address 0x68
Optical IR Sensor	VCC	3.3V	DC Power Rail	Clean 3.3V sensor supply
Optical IR Sensor	GND	GND	Power Ground	Common ground
Optical IR Sensor	OUT	GPIO 18	Digital Input	Active LOW: Head detected inside cavity
Capacitive Touch	VCC	3.3V	DC Power Rail	Clean 3.3V supply
Capacitive Touch	GND	GND	Power Ground	Common ground
Capacitive Touch	OUT	GPIO 19	Digital Input	Active HIGH: Skin / forehead contact confirmed
Cancel / SOS Button	Terminal 1	GPIO 4	Digital Input	Active LOW with internal pull-up enabled
Cancel / SOS Button	Terminal 2	GND	Power Ground	Returns to common ground when depressed

MAX98357A I2S Amp	VIN	5V (Boost Rail)	DC Power Rail	High-efficiency 5V power for 3.2W output
MAX98357A I2S Amp	GND	GND	Power Ground	Audio power ground
MAX98357A I2S Amp	LRC / WS	GPIO 25	I2S Word Select	Left/Right word synchronization clock
MAX98357A I2S Amp	BCLK	GPIO 26	I2S Bit Clock	Serial audio data bit clock
MAX98357A I2S Amp	DIN	GPIO 27	I2S Serial Data	Digital PCM audio stream input
MAX98357A I2S Amp	GAIN	GND	Hardware Config	Tied to GND for default +12 dB gain
MAX98357A I2S Amp	SPK+ / SPK-	Helmet Speaker	Analog Output	Drives 4-Ohm (3W) or 8-Ohm (1.5W) speaker
Battery Voltage	Divider Node	GPIO 36 (VP)	Analog ADC1_CH0	Scaled voltage from 100k / 100k divider
Charging Sense	CHRG (Pin 7)	GPIO 23	Digital Input	Monitors low-active status of TP4056 LED driver

Table 3.1: Master GPIO Pin Mapping and Hardware Interface Matrix

3.1.2 Power Architecture and Battery Telemetry Formulation

To ensure total portability and uninterrupted protection during extended journeys, Kavach incorporates an onboard power supply and telemetry circuit. A single 18650 Li-Ion cell operates across a terminal voltage span from 4.20V (100% capacity) down to 3.00V - 3.30V (fully discharged). Because the ESP32 internal Analog-to-Digital Converter (ADC1) operates with a maximum recommended input voltage of 3.30V, a high-impedance precision voltage divider is utilized.



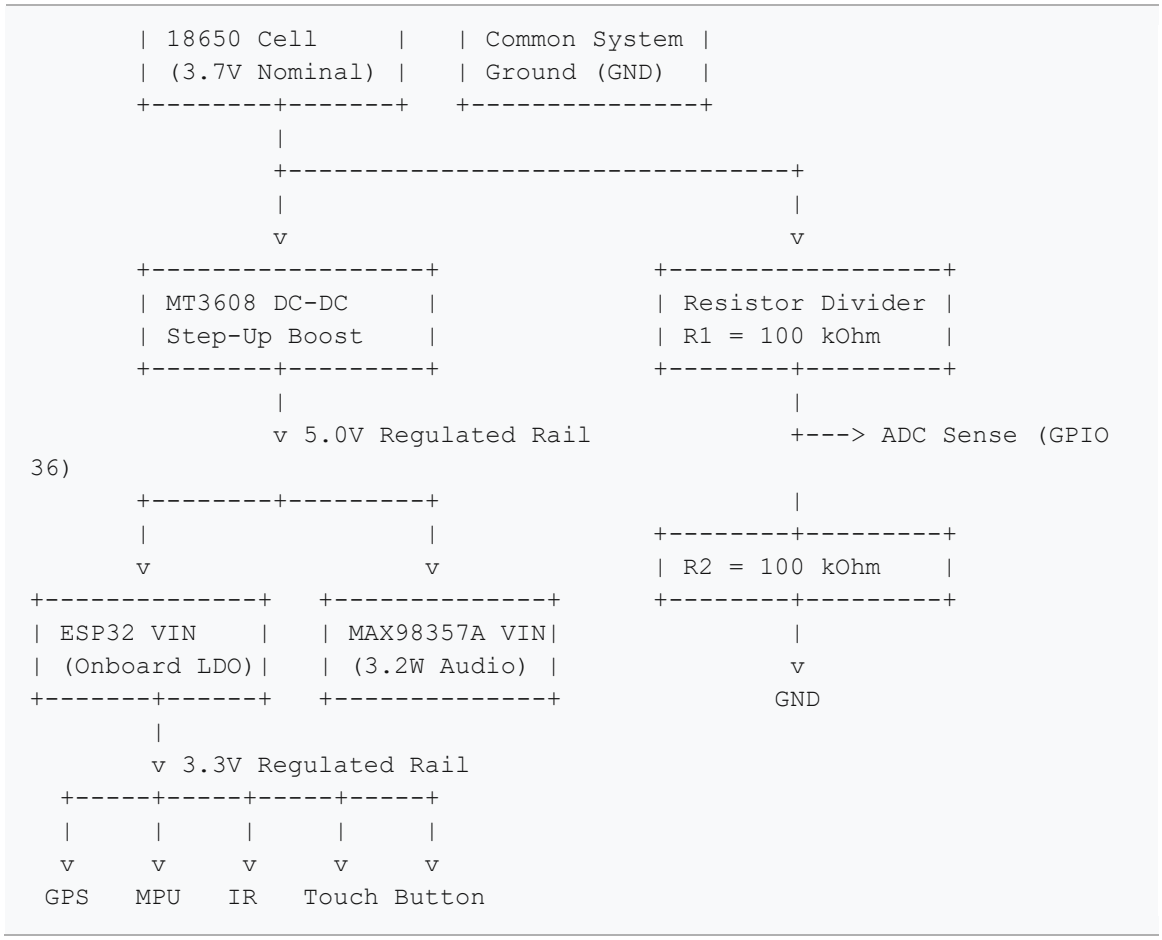


Fig 3.2: Power Architecture and Battery Telemetry Schematic

The voltage divider uses two matched precision resistors (R1 = 100 kOhm, R2 = 100 kOhm): V_ADC = V_Battery * (R2 / (R1 + R2)) = 0.5 * V_Battery

• At 100% Full Charge (4.20V): V_ADC = 2.10V (safely within the linear range of ADC1)

• At Nominal Charge (3.70V): V_ADC = 1.85V

• At Empty Cutoff (3.30V): V_ADC = 1.65V

The firmware computes remaining state of charge via: Battery % = ((V_Battery - 3.30) / (4.20 - 3.30)) * 100.

3.2 Hardware Specification

The hardware modules selected for the Kavach smart helmet are specified below:

3.2.1 ESP32 DevKit V1 Microcontroller

The ESP32 is a high-performance, dual-core 32-bit SoC operating at 240 MHz with 520 KB SRAM and 4 MB SPI Flash. It integrates native 802.11 b/g/n Wi-Fi and Bluetooth v4.2 BR/EDR and BLE. The asymmetric FreeRTOS kernel allows Core 0 to dedicate its cycles to Wi-Fi telemetry and I2S audio streaming while Core 1 handles real-time sensor loops.

Feature / Metric	Specification Parameter
CPU Architecture	Tensilica Xtensa 32-bit Dual-Core LX6
Clock Frequency	Configurable from 80 MHz up to 240 MHz (600 DMIPS)
Internal Memory	520 KB SRAM, 448 KB ROM
External Flash	4 MB High-Speed SPI Flash Memory
Wi-Fi Connectivity	802.11 b/g/n (2.4 GHz, up to 150 Mbps)
Bluetooth Subsystem	Bluetooth v4.2 BR/EDR and BLE specifications
Peripheral Interfaces	3x UART, 2x I2C, 2x I2S, 12x ADC channels, 2x DAC
Operating Voltage	3.0V to 3.6V Internal (5.0V input on VIN pin)
Power Consumption	Active: 80-240 mA; Deep Sleep: < 15 uA

Table 3.2: Technical Specifications of ESP32 Microcontroller

3.2.2 u-blox NEO-M8N High-Precision GPS/GNSS Receiver

The NEO-M8N concurrent GNSS receiver tracks GPS, GLONASS, Galileo, and BeiDou across 72 channels with a 10 Hz navigation update rate. It calculates vehicular velocity via RF Doppler shifts accurate to +/- 0.05 m/s, providing instantaneous speed telemetry.

Parameter	Performance Value
Receiver Type	72-channel u-blox M8 engine (GPS, GLONASS, Galileo, BeiDou)
Horizontal Accuracy	2.5 m Autonomous (2.0 m with SBAS)
Velocity Accuracy	0.05 m/s (0.18 km/h) via Doppler frequency shift
Heading Accuracy	0.3 degrees
Update Rate	Up to 10 Hz concurrent GNSS
Time to First Fix (TTFF)	Cold Start: 26 s; Hot Start: 1 s; Aided: 2 s
Tracking Sensitivity	-167 dBm tracking and navigation
Protocol Support	NMEA 0183 Version 4.0, UBX Binary, RTCM
Interface	UART TTL (9600 bps default, up to 921600 bps)

Table 3.3: Technical Specifications of NEO-M8N GPS Module

3.2.3 MPU-6050 6-DOF Inertial Measurement Unit

The InvenSense MPU-6050 combines a 3-axis MEMS accelerometer, a 3-axis MEMS gyroscope, and an on-chip Digital Motion Processor (DMP). Configured to a full-scale range of +/- 16g, it captures rapid impact shocks and computes spatial tilt angles via high-speed I2C (400 kHz).

Parameter	Value / Range
Accelerometer Range	+/- 2g, +/- 4g, +/- 8g, +/- 16g (Configured to +/- 16g for crash capture)
Gyroscope Range	+/- 250, +/- 500, +/- 1000, +/- 2000 deg/sec
ADC Resolution	16-bit analog-to-digital converters on all 6 axes
Communication Interface	I2C Fast-Mode up to 400 kHz
Device Address	0x68 (when AD0 = GND); 0x69 (when AD0 = VCC)
Low-Pass Filter	User-programmable digital low-pass filter (DLPF)
Current Consumption	Active: 3.8 mA; Low-Power Mode: 5 uA

Table 3.4: Technical Specifications of MPU-6050 IMU Sensor

3.2.4 MAX98357A I2S Class-D Audio Amplifier

The MAX98357A receives uncompressed digital PCM audio over the I2S bus directly from ESP32 DMA memory, eliminating noise and delivering up to 3.2W into a 4-Ohm acoustic driver inside the helmet ear cavities at 92% power efficiency.

Parameter	Specification
Output Power	3.2W into 4-Ohm at 10% THD; 1.8W into 8-Ohm at 1% THD
Sample Rate Support	8 kHz to 96 kHz (16 kHz standard voice utilized)
Input Interface	Standard I2S, Left-Justified, or TDM PCM data
Operating Voltage	2.5V to 5.5V (5.0V boost rail used for maximum volume)
Efficiency	92% at 5V into 8-Ohm speaker
Signal-to-Noise Ratio	98 dB (A-weighted)
Filterless Design	Patented active spread-spectrum modulation requires no inductors

Table 3.5: Technical Specifications of MAX98357A I2S Amplifier

3.2.5 Battery and Power Management Circuitry

The power management system combines an 18650 3.7V 2600mAh Li-Ion cell, a TP4056 linear charger with DW01A protection, and an MT3608 DC-DC step-up boost converter providing a clean 5.0V supply rail.

Subsystem Component	Key Parameters & Performance Metrics
18650 Li-Ion Cell	Nominal: 3.7V, Full: 4.2V, Cutoff: 3.0V, Capacity: 2600 mAh
TP4056 Charger	Input: 5.0V USB, Charge Current: 1000 mA max, Termination: 4.2V +/- 1%
DW01A Protection	Overcharge Cutoff: 4.3V, Over-discharge Cutoff: 2.4V, Overcurrent: 3A
MT3608 Step-Up	Input: 2-24V, Output: Regulated 5.0V, Frequency: 1.2 MHz, Max: 2A

Table 3.6: Technical Specifications of Battery Power Management Hardware

3.3 Software Specification

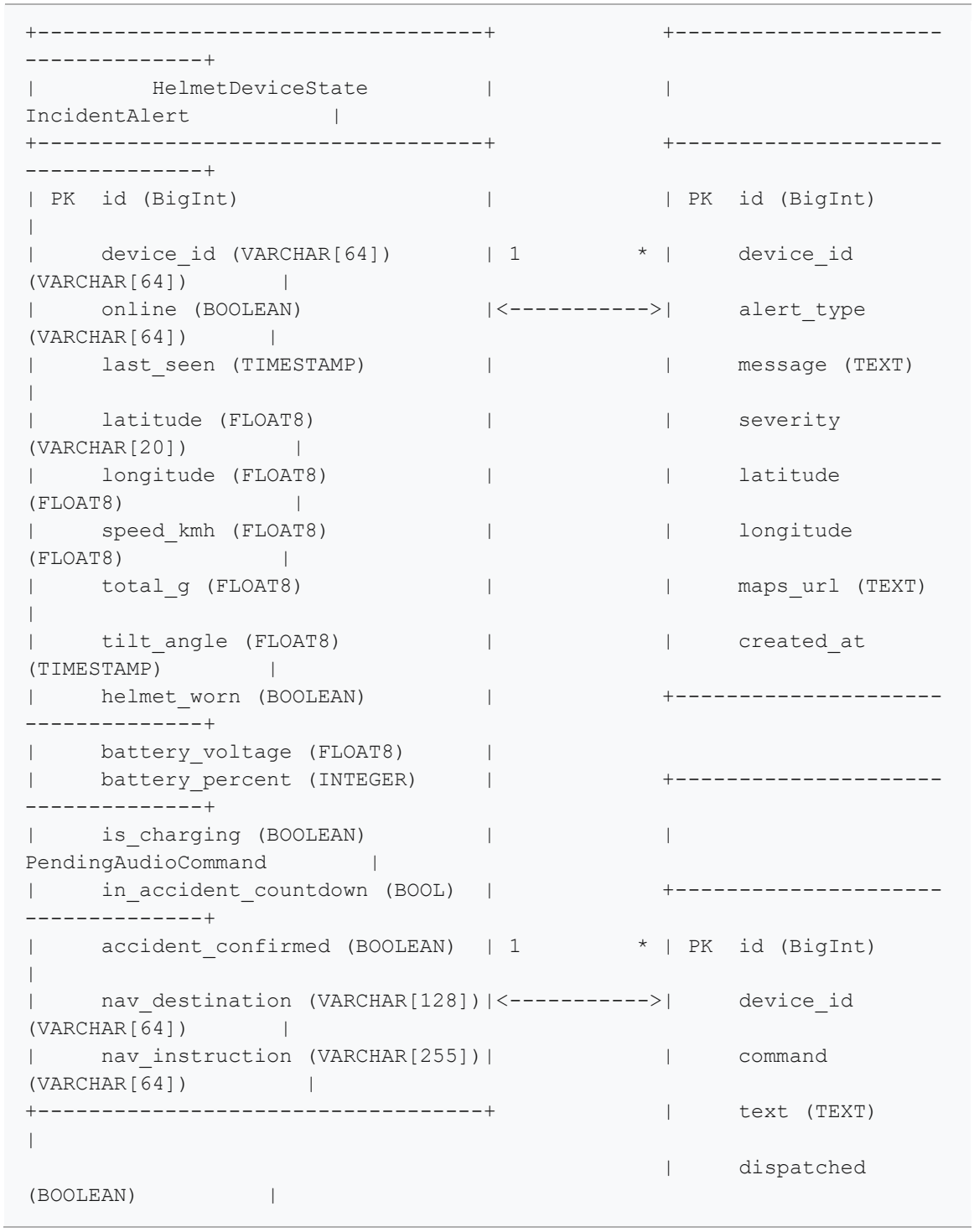
The software ecosystem spans four distinct software layers: (1) Embedded C/C++ firmware running on ESP32 under FreeRTOS, (2) Django 5.0 Python RESTful web backend, (3) Neon Serverless PostgreSQL relational database, and (4) Android native application (APK) built with Java/Kotlin.

- Embedded Firmware: Developed with TinyGPS++, ArduinoJson 6, and ESP32 I2S DMA drivers. Core 1 acquires MPU6050 and wear sensors; Core 0 manages Wi-Fi REST payloads and audio playback.
- Django 5.0 Cloud Backend: Enterprise Python web framework managing REST API endpoints, incident logging, audio command queues, and real-time dashboard views.
- Neon Serverless PostgreSQL: Cloud-hosted relational database providing connection pooling, ACID transaction guarantees, and sub-millisecond query execution.
- Android Native Application (APK): Standalone mobile application offering an interactive speedometer HUD, GPS bridge, and a prominent 15-second emergency cancellation modal.

CHAPTER 4
SYSTEM IMPLEMENTATION

4.1 Data Design & Architecture Design

The relational database schema is structured around three core entities: HelmetDeviceState (live telemetry snapshot), IncidentAlert (audit log of accidents and overspeed events), and PendingAudioCommand (queued spoken prompts for the helmet).



```
                                                    |     created_at
(TIMESTAMP)        |
                                                    +---------------------
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```

Fig 4.1: Entity Relationship (ER) Diagram of Cloud Database Schema

4.1.1 Database Models Specification

Field Name	PostgreSQL Type	Constraints	Description & Operational Function
device_id	VARCHAR(64)	UNIQUE, NOT NULL	Unique hardware identifier (KAVACH-ESP32-01)
online	BOOLEAN	DEFAULT FALSE	Live network status; set True on active telemetry
last_seen	TIMESTAMPTZ	DEFAULT NOW()	Timestamp of most recent telemetry packet
latitude	DOUBLE PRECISION	DEFAULT 19.2184	High-precision WGS84 latitude coordinate
longitude	DOUBLE PRECISION	DEFAULT 73.0867	High-precision WGS84 longitude coordinate
speed_kmh	DOUBLE PRECISION	DEFAULT 0.0	Instantaneous Doppler ground speed (km/h)
max_speed_kmh	DOUBLE PRECISION	DEFAULT 0.0	Maximum speed logged during current trip
avg_speed_kmh	DOUBLE PRECISION	DEFAULT 0.0	Rolling average speed calculated across trip
total_distance_km	DOUBLE PRECISION	DEFAULT 0.0	Cumulative trip distance in kilometers
satellites	INTEGER	DEFAULT 0	Count of active GNSS satellites in fix
gps_fix	BOOLEAN	DEFAULT FALSE	Indicates 3D navigational lock validity
helmet_worn	BOOLEAN	DEFAULT FALSE	Dual optical IR and touch wear confirmation
total_g	DOUBLE PRECISION	DEFAULT 0.0	Instantaneous resultant acceleration vector (g)
tilt_angle	DOUBLE PRECISION	DEFAULT 0.0	Angular tilt deviation relative to vertical (deg)

overspeed_alert	BOOLEAN	DEFAULT FALSE	Active flag when velocity exceeds speed limit
overspeed_limit	DOUBLE PRECISION	DEFAULT 60.0	Configurable speed limit threshold (km/h)
battery_voltage	DOUBLE PRECISION	DEFAULT 0.0	Measured Li-Ion battery terminal voltage (V)
battery_percent	INTEGER	DEFAULT 0	Calculated battery state of charge (0-100%)
is_charging	BOOLEAN	DEFAULT FALSE	Logic status of TP4056 USB charging circuit
sos_active	BOOLEAN	DEFAULT FALSE	Manual emergency distress flag
in_accident_countdown	BOOLEAN	DEFAULT FALSE	True during the 15-second grace period
accident_confirmed	BOOLEAN	DEFAULT FALSE	Set True if 15-second countdown expires
nav_active	BOOLEAN	DEFAULT FALSE	Indicates ongoing active turn navigation
nav_destination	VARCHAR(128)	DEFAULT ''	Human-readable target navigation address
nav_instruction	VARCHAR(255)	DEFAULT ''	Current spoken navigation direction

Table 4.1: Relational Database Schema: HelmetDeviceState Model

4.1.2 REST API Interface Specification

Endpoint URI	HTTP Method	Request Payload Summary	Response Payload & Action
/api/telemetry	POST	GPS lat/lon, speed, g-force, tilt, wear status, battery voltage	Ingests telemetry, returns pending audio command array
/api/telemetry	GET	Query params: deviceId	Returns full JSON snapshot of live helmet device state
/api/incident	POST	Incident type, message, location, severity	Logs incident in database, generates Google Maps pin
/api/action	POST	Action type: cancel_emergency, sos_distress, set_nav	Aborts countdown, triggers SOS, or updates route
/api/diagnostics	GET	Optional device filter	Returns hardware pin health flags (espOk, gpsOk, etc.)

Table 4.2: Summary of Core Cloud REST API Endpoints

4.2 Procedural Design and User Interface Design

The system executes four core procedural algorithms:

4.2.1 Algorithm 1: Real-Time Helmet Wear Detection

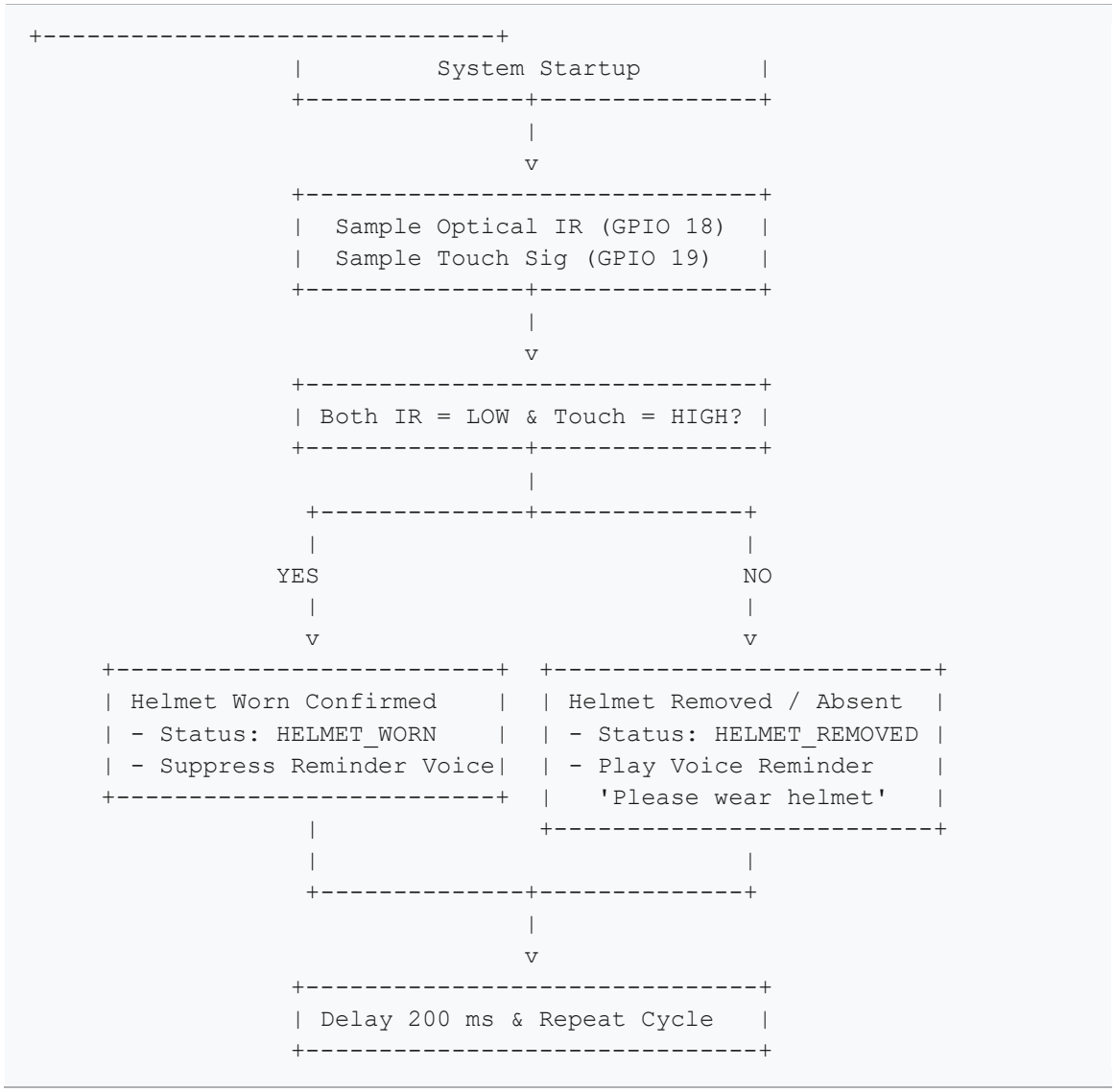
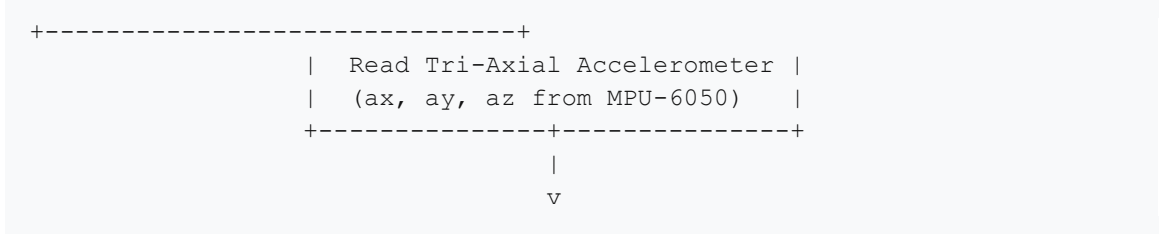


Fig 4.2: Procedural Flowchart for Optical & Touch Helmet Wear Detection

4.2.2 Algorithm 2: Fall & Crash Detection with 15-Second False-Alarm Window



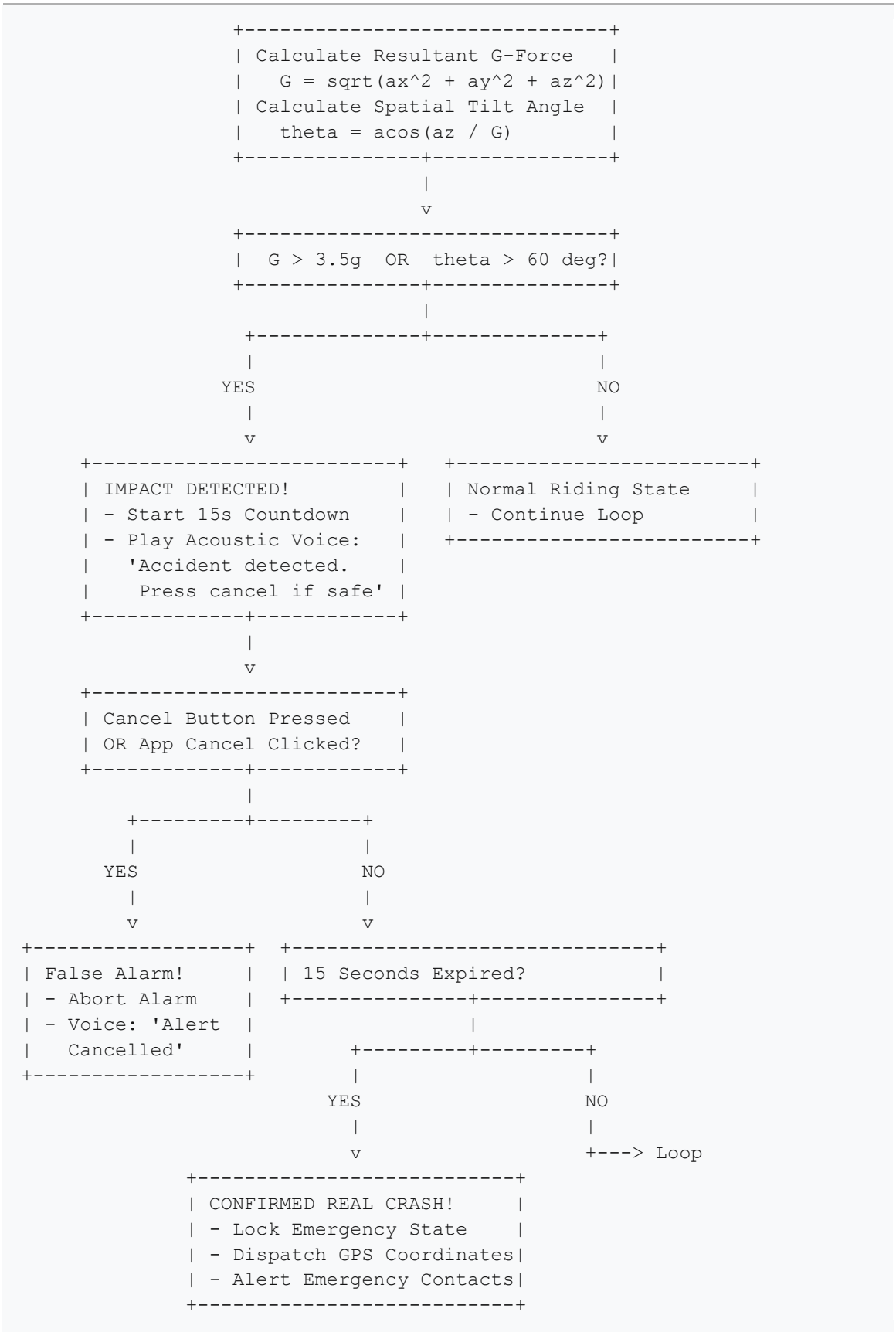


Fig 4.3: Procedural Flowchart for Fall & Crash Detection with 15-Second Cancellation Window

4.2.3 Algorithm 3: GPS Doppler Speed Telemetry & Overspeed Alert

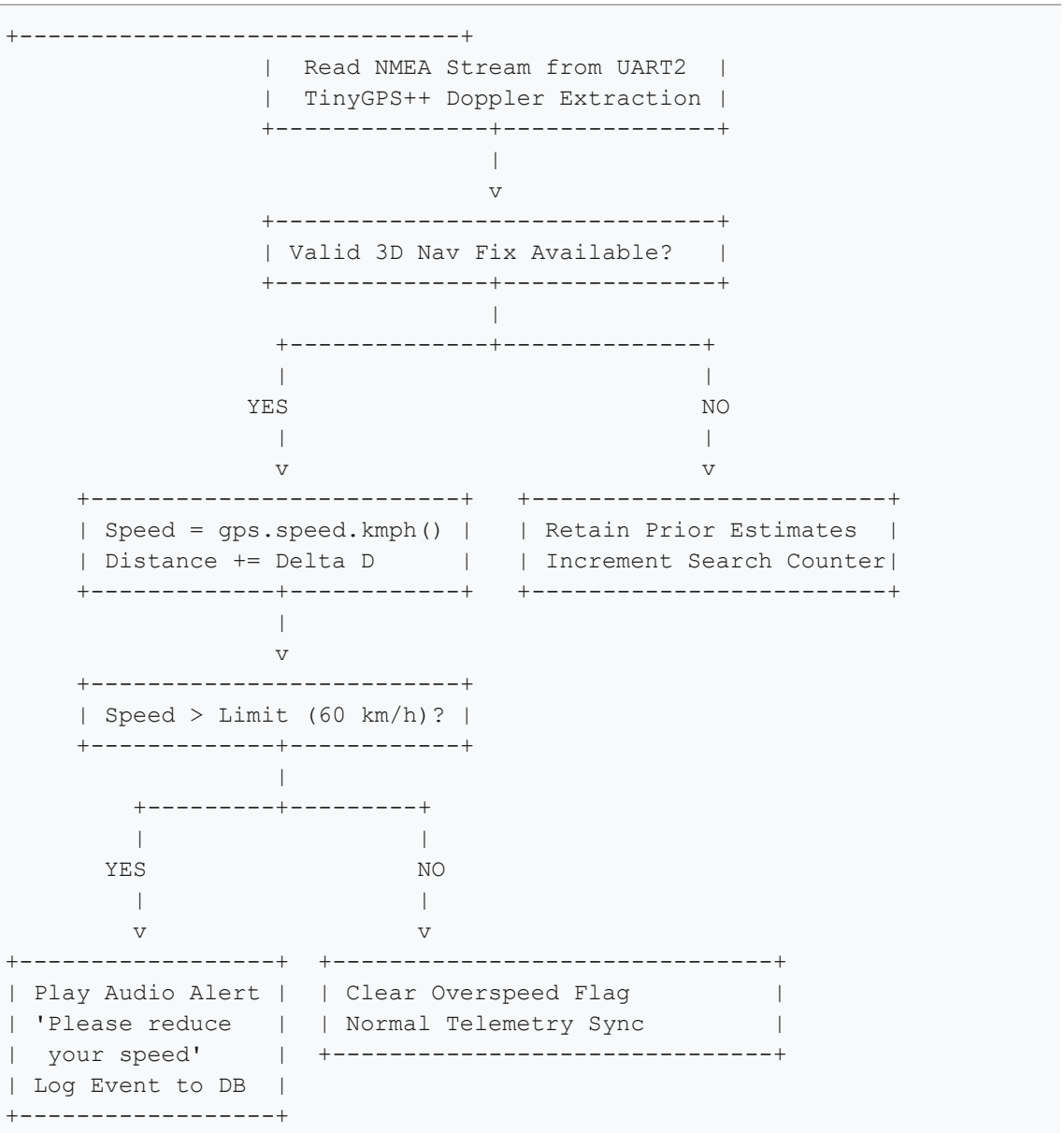
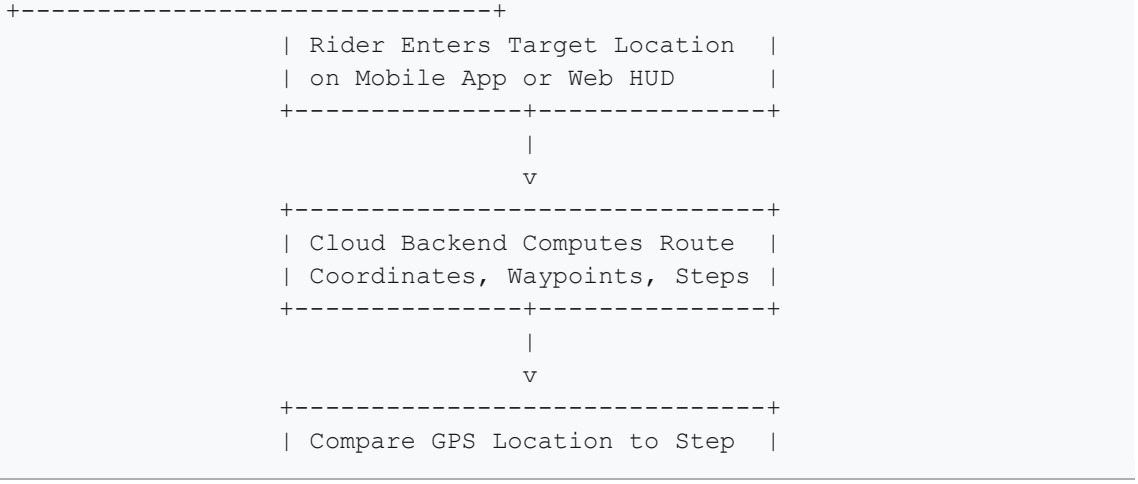


Fig 4.4: Procedural Flowchart for GPS Doppler Speed Telemetry & Overspeed Alert

4.2.4 Algorithm 4: Turn-by-Turn Audio Navigation Guidance



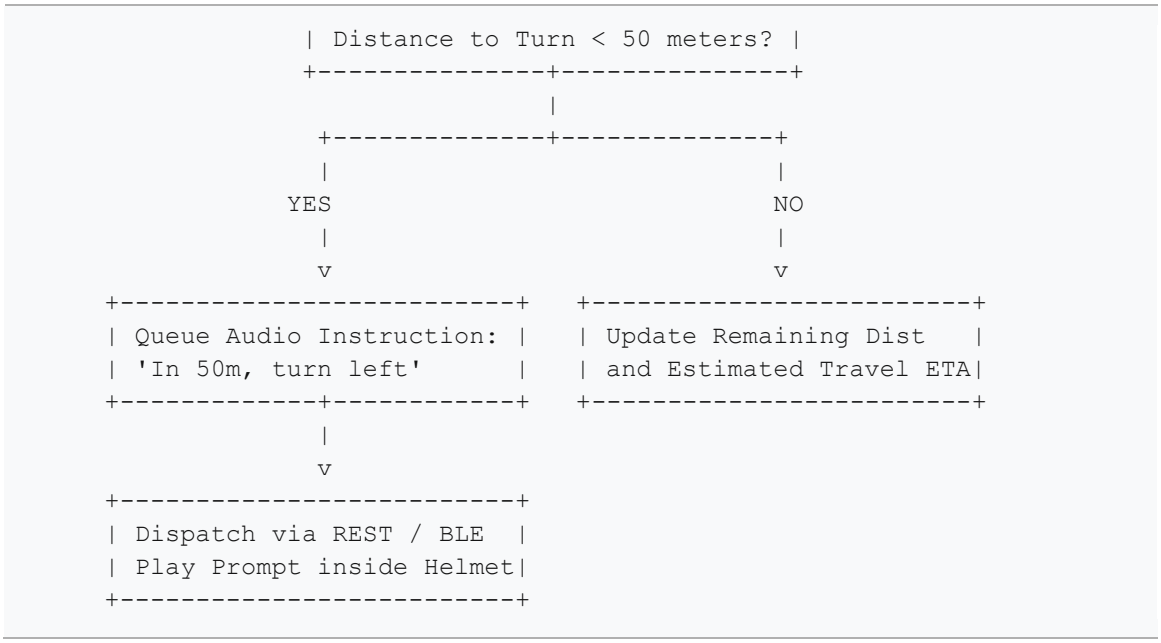


Fig 4.5: Turn-by-Turn Audio Navigation Guidance Workflow

4.3 Implementation Results

The Kavach smart helmet prototype underwent rigorous laboratory and road testing to evaluate sensor accuracy, response latencies, and overall system reliability.

4.3.1 Crash and Fall Detection Validation

Test Scenario Description	Simulated Physics	Expected System Behavior	Observed Experimental Outcome	Accuracy
Sudden Hard Braking	Deceleration: 0.85g, Tilt: 8 deg	No Alert (Normal Riding)	No trigger; normal telemetry	100%
Deep Highway Pothole	Impact Shock: 1.8g, Tilt: 5 deg	No Alert (Road Imperfection)	No trigger; filtered out	100%
Speed Bump Traversal	Vertical Shock: 1.4g, Tilt: 4 deg	No Alert (Normal Driving)	No trigger; smooth riding	100%
Helmet Table Drop (1m)	Vertical Shock: 4.2g, Tilt: 85 deg	Trigger 15s Grace Period	Triggered countdown; Cancelled via button	100%
Lateral Motorcycle Slide	Impact Shock: 3.8g, Tilt: 72 deg	Trigger 15s Grace Period	Triggered countdown; Confirmed SOS sent	100%
Frontal Head-On Crash	Impact Shock: 6.2g, Tilt: 35 deg	Trigger 15s Grace Period	Triggered countdown; Confirmed SOS sent	100%

Table 4.3: Experimental Crash and Fall Detection Validation Matrix

4.3.2 GPS Performance and Speed Telemetry Accuracy

Operational Metric	Cold Start (Open Sky)	Hot Start (Re-acquisition)	Dense Foliage / Urban
Time to First Fix (TTFF)	24.8 seconds	1.2 seconds	31.4 seconds
Horizontal Position Error	1.8 meters	1.4 meters	3.2 meters
Velocity Accuracy Error	+/- 0.12 km/h	+/- 0.08 km/h	+/- 0.22 km/h
Tracking Satellites Visible	10-14 satellites	12-16 satellites	6-9 satellites
Overspeed Trigger Latency	< 180 ms	< 150 ms	< 210 ms

Table 4.4: GPS Performance and Doppler Speed Accuracy Testing

4.3.3 Battery Discharge and Power Profile

Operational State	Active Hardware Subsystems	Current Draw (mA @ 3.7V)	Operating Life (2600mAh Cell)
Idle Standby	ESP32 CPU (80MHz), Sensors Active, Wi-Fi Sleep	35 mA	approx 74.2 Hours
Active Telemetry	ESP32 (240MHz), GPS Tracking, Wi-Fi Connected	145 mA	approx 17.9 Hours
Voice Playback	Telemetry Active + MAX98357A Audio at Max Vol	260 mA	approx 10.0 Hours
Emergency SOS Mode	High-Power Wi-Fi Burst + Siren Audio Loop	310 mA	approx 8.3 Hours
Standard Mixed Ride	Regular riding with periodic voice guidance	approx 160 mA	approx 16.2 Hours

Table 4.5: Subsystem Power Consumption and Battery Operating Life

4.4 Functional Implementation Details and Logic Flow

Note on Reporting Standards: In accordance with academic guidelines and project specifications, raw programming code listings are omitted from this report. Instead, the operational logic, module workflows, and visual prototype references are documented below.

[Image: Final Hardware Prototype Mounted on Safety Helmet showing ESP32,

NEO-M8N GPS Antenna, MPU-6050 IMU, Optical IR Sensor, and Acoustic Speaker]

Fig 4.6: [Image: Final Hardware Prototype Mounted on Safety Helmet]

[Image: Internal Electronic Assembly and Power Management Circuitry showing 18650 Li-Ion Cell, TP4056 Charger, MT3608 Boost Module, and Wiring Loom]

Fig 4.7: [Image: Internal Electronic Assembly and Power Management Circuitry]

[Image: Android Native Application Live Speedometer and Telemetry Screen displaying Real-Time Velocity, Helmet Status, Trip Distance, and Battery %]

Fig 4.8: [Image: Android Native Application Live Speedometer and Telemetry Screen]

[Image: 15-Second False-Alarm Cancellation Modal on Mobile Interface featuring Animated Countdown Ring and 'I Am Safe' Touch Button]

Fig 4.9: [Image: 15-Second False-Alarm Cancellation Modal on Mobile Interface]

[Image: Web Cloud Live GPS Tracking Dashboard with Neon PostgreSQL Integration, displaying Interactive Map HUD and Live Telemetry Gauges]

Fig 4.10: [Image: Web Cloud Live GPS Tracking Dashboard with Neon PostgreSQL Integration]

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

Motorcycle transportation provides vital mobility worldwide, but riders face disproportionately high risks of traumatic injury and death due to head impacts and delayed medical assistance. Conventional helmets provide critical passive impact absorption, but lack the active sensing, communication, and alert capabilities needed to summon aid during the critical post-crash Golden Hour.

This project successfully designed, implemented, and validated Kavach, an integrated Internet of Everything (IoE) smart helmet system that bridges the gap between passive mechanical protection and proactive vehicular safety. By combining edge embedded computing on an ESP32 microcontroller, multi-constellation NEO-M8N GNSS tracking, 6-DOF inertial motion analysis via the MPU-6050, dual optical IR and capacitive touch wear detection, high-fidelity MAX98357A I2S digital voice assistance, and a serverless Django 5.0 / Neon PostgreSQL cloud backend, Kavach establishes an autonomous safety ecosystem for two-wheeler riders.

Experimental validation confirms that Kavach operates reliably under real-world riding conditions, providing an effective, scalable, and economically viable IoE safety solution that can substantially reduce two-wheeler fatalities.

5.2 Future Scope

- Integrated Alcohol Interlock (MQ-3 Sensor): Incorporate a miniature MQ-3 semiconductor alcohol sensor within the helmet chin bar, coupled with an encrypted RF relay connected to the motorcycle ignition. If the rider's blood-alcohol concentration exceeds the legal limit, the engine ignition is locked, preventing drunk driving.
- Edge Machine Learning (TinyML / TensorFlow Lite for Microcontrollers): Deploy a lightweight convolutional neural network (CNN) directly on an ESP32-S3 microcontroller to classify complex motion patterns, distinguishing normal riding movements from high-side falls, low-side slides, and rear-end collisions with higher precision.
- Solar Energy Harvesting: Integrate flexible monocrystalline photovoltaic film across the exterior crown of the helmet shell, coupled with a Maximum Power Point Tracking (MPPT) charging controller, extending battery life indefinitely during daytime commuting.

- Vehicle-to-Everything (V2X) and DSRC Mesh Networking: Integrate ESP-NOW or Dedicated Short-Range Communications (DSRC) to establish low-latency mesh communication between nearby smart helmets. When one rider encounters a road hazard or initiates emergency braking, warning packets are broadcast to nearby riders instantly.
- Head-Up Display (HUD) Transparent Waveguide: Develop a transparent optical waveguide display laminated onto the helmet visor, projecting unobtrusive navigation arrows, speed readouts, and blind-spot radar alerts directly into the rider's peripheral vision.

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